



# OncoBind

Structure-based decision support for precision oncology — helping physicians choose the targeted therapy the biology actually supports.

# Physicians choosing between targeted therapies are flying blind on the biology.

- Molecular subtyping now routinely identifies the exact mutated target driving a patient's cancer — but for most such targets there are now **three to five FDA-approved drugs**, not one.
- Selection among them is guided largely by approval recency, clinician familiarity, and label-reading — **not by comparative structural or binding data**, which is scattered across separate primary-literature sources for each drug.
- Adverse-effect tradeoffs — a black-box cardiovascular warning versus a milder but less potent alternative — are buried in **separate, non-comparative prescribing information** documents.

5 major kinase-target families. 17+ approved drugs across them. No structure-based comparison tool built for the point of care — until now.

# Structure-based drug comparison just became fast enough for clinical workflows.

01

## Models got fast

AlphaFold-era docking and affinity-prediction models (DiffDock, Boltz2, and peers) now return a predicted binding pose and affinity in seconds to minutes — down from weeks of wet-lab crystallography.



02

## Models became API-accessible

Hosted inference platforms (NVIDIA BioNeMo among others) put these models behind a simple API — no in-house computational chemistry team or GPU cluster required to use them.



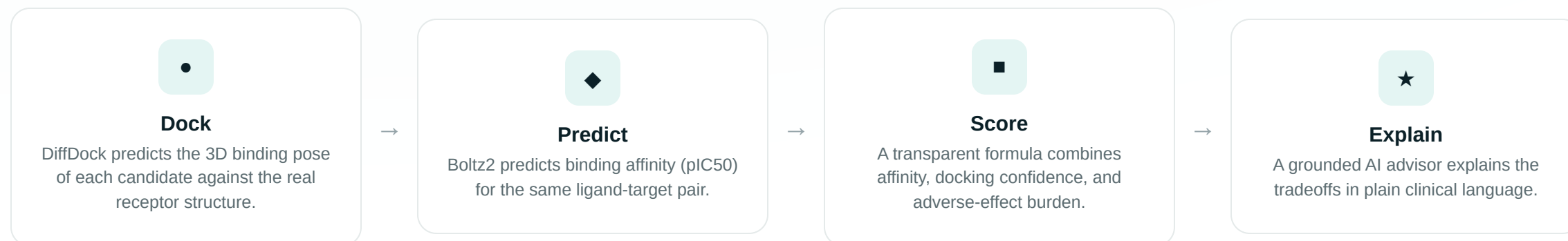
03

## The drug landscape exploded

Targeted-therapy approvals are accelerating — more kinase-inhibitor approvals in the last five years than the prior fifteen, each adding to the comparison problem.

# OncoBind turns each targeted-therapy decision into a structural comparison, not a memory test.

For a diagnosed molecular subtype, OncoBind docks every approved candidate drug against the real target structure, predicts binding affinity, and combines it — transparently — with known adverse-effect burden into one comparative, explainable score.



# Five curated cancer scenarios, ready to explore today



**OncoBind**  
Precision Oncology Therapy Advisor

built on  Base44

## MOLECULAR TARGET · THERAPY COMPARISON

### Choose the therapy the biology actually supports.

For each molecularly-defined cancer, OncoBind docks every approved candidate drug against the real target structure, predicts binding affinity, and weighs the result against known adverse-effect burden — so the tradeoffs are visible, not buried in prescribing information.

 **Research & education tool only.** Binding poses and affinities are computational predictions, not clinical measurements. This is not a diagnostic or prescribing device and does not replace clinical judgment, current FDA prescribing information, or institutional guidelines.



NON-SMALL CELL LUNG CANCER (NSCLC)



#### EGFR-Mutant Non-Small Cell Lung Cancer

EGFR exon 19 deletion / L858R activating mutations

Target: EGFR Tyrosine Kinase Domain · PDB 1M17



CUTANEOUS MELANOMA



#### BRAF V600E-Mutant Melanoma

BRAF V600E activating mutation

Target: BRAF V600E Kinase Domain · PDB 3OG7

EGFR-mutant NSCLC, BRAF V600E melanoma, HER2+ breast cancer, CML, and ALK+ NSCLC — each backed by a real crystallized target structure and its full slate of approved therapies.

# Every score is explainable — down to its three inputs and their weights



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← All scenarios

CUTANEOUS MELANOMA

**BRAF V600E-Mutant Melanoma**

BRAF V600E activating mutation

**BRAF V600E KINASE DOMAIN · PDB 3OG7 · UNIPROT P15056**  
The V600E mutation locks BRAF's activation loop in an active conformation, driving constitutive MAPK pathway signaling independent of upstream RAS activation. BRAF inhibitors selectively bind the mutant active-conformation ATP pocket.

| DRUG                                   | CLASS                                     | DOCKING<br>CONF. | PIC50 | COMPOSITE<br>SCORE       | KEY ADVERSE EFFECTS                                                |
|----------------------------------------|-------------------------------------------|------------------|-------|--------------------------|--------------------------------------------------------------------|
| Encorafenib<br><small>Braftovi</small> | Selective<br>BRAF<br>V600E/K<br>inhibitor | 0.12             | 11.47 | <div><div></div>64</div> | <div>Nausea</div> <div>Palmar-plantar<br/>erythrodysesthesia</div> |
|                                        | 1st-                                      |                  |       |                          |                                                                    |

 **Oncology Advisor**

Ask about mechanism of action, binding predictions, or adverse-effect tradeoffs for BRAF V600E-Mutant Melanoma.

Physicians can ask the grounded AI advisor why a drug scored the way it did — it only cites data present in the app, never an invented number.

# No black box. Every number traces back to a named model or a public database.

Unlike opaque "AI risk score" tools, OncoBind's composite score is a published, auditable formula — the same one the in-app AI advisor is instructed to explain verbatim on request.

```
// composite_score, normalized within each scenario's candidate drugs
composite_score = 100 × clamp(
  0.45 × binding_affinity_norm
  + 0.35 × docking_confidence_norm
  + 0.20 × (1 - adverse_effect_burden_norm)
)
```

# Every molecular tumor board is a potential user.

- Targeted therapy is the fastest-growing category of oncology drug approvals, and each new approval into an existing target class **adds to the comparison problem** OncoBind solves — the market grows with the pipeline it depends on.
- Initial buyers: academic medical centers and health systems running molecular tumor boards and precision-oncology programs, where this exact multi-drug comparison conversation already happens manually today.
- Adjacent buyers: pharma medical-affairs and market-access teams evaluating competitive binding/selectivity positioning pre- and post-launch.

## PLACEHOLDER — NEEDS SOURCED DATA

Quantified TAM/SAM/SOM sizing to be added from a licensed market-research source before external distribution of this deck.



# Three paths to revenue, sequenced by validation risk

## NEAR-TERM

### Institutional SaaS

Per-site or per-seat subscription for health systems and oncology practices with molecular tumor boards.

## MID-TERM

### Data & API licensing

Structured binding/affinity comparison data licensed to pharma medical-affairs and payer/formulary teams.

## LONG-TERM

### EHR-embedded CDS

Integration into oncology EHR workflows as point-of-care clinical decision support.

## PLACEHOLDER — NEEDS COMMERCIAL VALIDATION

Pricing, initial design-partner pipeline, and current traction to be added once pilot conversations are underway.

# Structural, comparative, and transparent — a gap the incumbents don't fill

| CATEGORY                                                    | STRENGTH                                                   | GAP ONCOBIND FILLS                                                                |
|-------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Static reference tools</b><br>(NCCN, Lexicomp, UpToDate) | Comprehensive, trusted, guideline-anchored                 | Not structural or comparative — read one drug at a time, no binding-based ranking |
| <b>OncoBind</b>                                             | Structure-based, transparent, explainable per-drug scoring | —                                                                                 |
| <b>Black-box AI risk tools</b>                              | Data-driven, can flag risk patterns                        | Opaque scoring, heavy regulatory burden, no grounding in physical binding data    |

# Built as decision support, by design — not a diagnostic device.

## Human-in-the-loop, always

OncoBind never recommends a specific prescription. Every view carries a persistent disclaimer, and the AI advisor is instructed to frame findings as considerations for the treating physician's judgment.

## Positioned as research/education software today

Current scope is explicitly a research and education decision-support tool. Binding predictions are computational, not clinically validated measurements.

## Auditable by construction

Every displayed number cites its source (RCSB PDB, UniProt, PubChem, or the named prediction model) — nothing is a hidden training-data artifact.

## Clear path forward

Should the product move toward higher-stakes clinical use, its architecture — transparent scoring, sourced data, no hidden model — is the right starting point for eventual regulatory engagement.

# From 5 scenarios to a platform

## NEAR-TERM

### Expand scenario & drug coverage

Grow beyond the initial 5 scenarios / 17 drugs to the full set of clinically relevant targeted-therapy classes; add on-demand live docking for investigational or off-label candidates.

## NEAR-TERM

### Pilot with a design-partner health system

Validate the workflow and scoring methodology directly with a molecular tumor board.

## LONG-TERM

### Broader pharmacogenomic decision support

Extend beyond oncology-target binding into patient-specific pharmacogenomic interaction data.

## EXPLORATORY

### Early-detection research collaborations

The same binding-prediction engine could, in principle, inform future biomarker or early-detection research — this is a speculative research direction, not a current product capability.

# The people and the ask

**PLACEHOLDER**

**Founding team**

Add founder bios, relevant oncology/ML/clinical-software background, and advisors here.

**PLACEHOLDER**

**Current traction**

Add pilot conversations, LOIs, or design-partner status here once available.

**PLACEHOLDER**

**The ask**

Add raise amount, instrument, and target close date here.

**PLACEHOLDER**

**Use of funds**

Add allocation across engineering, clinical validation, and go-to-market here.

This deck describes a working technical prototype (live product linked below). Business sections marked "Placeholder" intentionally contain no fabricated figures — fill in with real, current data before sharing externally.



# Let's build the future of precision therapy selection.

[oncobind.app](https://oncobind.app) · [founder email placeholder]